Collaborative FMCW Radar Swarm with Onboard AI Decision Engine for Aerial Surveillance

Document 1 -- Proposal for M

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1. Collaborative FMCW Radar Swarm with Onboard AI for Aerial Surveillance

1.1 Problem Statement.

Contemporary aerial surveillance requires persistent, wide-area detection of low-observable targets at operationally sustainable cost. Single-platform airborne radar is vulnerable to attrition and cannot provide distributed coverage. Aran Technologies addresses this requirement through a five-drone swarm configured as a distributed FMCW radar -- a micro-AWACS architecture -- that distributes aperture across platforms, provides mission redundancy, and enables autonomous track management independent of continuous ground-station control.

1.2 Proposed Solution.

Aran Technologies proposes a five-hexacopter swarm, each platform carrying six 24 GHz FMCW radar panels (TI AWR1843, 60deg H-FOV per panel) providing full 360deg coverage per drone. A three-layer onboard AI pipeline autonomously processes raw radar data through to tactical decision output.

Layer 1 -- Signal Processing: AWR1843 DSP performs range-Doppler FFT and CFAR detection within 10 ms, outputting range, azimuth, radial velocity, and SNR.

Layer 2 -- Classification: Random Forest on Jetson classifies candidates as targets or clutter within 50 ms using SNR, velocity, range-rate, and RCS. Only confirmed detections reach Layer 3.

Layer 3 -- LLM Tactical Engine: Llama 3.2 3B (leader) and Gemma 2B (soldiers) receive JSON situation reports and generate tactical commands -- track reassignment, formation reallocation, sector reorientation, and threat alerts -- triggering only on Layer 2 confirmations.

1.3 Swarm Architecture and Resilience.

Command hierarchy: Ground Station > Leader Drone > Soldier autonomous mode. On leader heartbeat loss exceeding 2 seconds, the highest-battery soldier self-elects as leader via bully election protocol, assuming radar fusion and ASP publication within 2 seconds. On soldier loss, the leader LLM redistributes orphaned track IDs to the nearest active drone. Full Air Situation Picture continuity is sustained with three or more drones operational, satisfying the MBC-3 graceful degradation requirement.

A FastAPI Ground Control Station displays a consolidated real-time ASP at 2.5 Hz -- track table, polar radar display, leader identity, decision log, and timestamped JSON recording -- on a single browser screen with no external dependency.

1.4 Platform Specification.

Each hexacopter carries: TI AWR1843BOOST radar panels x6, indigenous STM32 flight controller (custom PCB), Doodle Labs AES-128 mesh radio, and Jetson edge compute (AGX Orin 64 GB on leader; Orin NX 16 GB on soldiers). AUW <= 4.3 kg. Endurance: ~32 min (6S, 10,000 mAh). Auto RTH on data-link loss or low battery. GNSS-denied resilience via EKF2 fusing optical flow, IMU, and barometer.

1.5 Indigenisation.

The complete intelligence layer -- CFAR software, Random Forest classifier, LLM tactical engine, ROS2 swarm coordination stack, GCS dashboard, custom antenna PCB design, and hexacopter airframe fabrication -- is 100% indigenously developed. Weighted indigenous content: >= 55% by mission-criticality, satisfying MBC-3 sec.2.25.

1.6 Phase I Deliverable.

Phase I complete: SITL simulation verified (ASP, rMADER trajectory deconfliction, LLM reallocation, graceful degradation), broadband and comms tested, single-drone and 5-drone swarm demo videos (1920x1080) ready -- live demonstration at New Delhi, 13-24 July 2026 (github.com/lekkalaharsha/aran-mbc3-swarm).